PHILIP NKWAM

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PROFILE

Computational Imaging researcher building imaging tools that work where scanners, bandwidth and expertise are scarce — and building the people who will use them. Selected from 350+ international applicants as the **first undergraduate and first African** to lead the Alzheimer's Association ISTAART Ambassador programme, directing 43 researchers across 35 countries, and one of two early-career representatives on the Executive Committee governing a 1,870-member neuroimaging community spanning 72 countries. Graduated **overall best in the research category** at the College of Medicine, University of Lagos. Work to date has put free tools and open data in the hands of African researchers: WebMRIQC removes the supercomputer requirement for MRI quality control & Cross-site harmonisation, and a MICCAI Open Data grant for an open annotated African mammography data. Founded Brain-AID as an undergraduate to build the neuroimaging training capacity his institution lacked — now 350+ members across 74 institutions and five countries.

LEADERSHIP & SCIENTIFIC GOVERNANCE

Global Lead Ambassador — Alzheimer's Association ISTAART **| Apr 2026 – Present**

- Selected from **350+ international applicants** as the first undergraduate and first African to lead the programme; direct **43 ambassadors across 35 countries**, the majority at doctoral or postdoctoral level.
- Coordinate scientific sessions at AAIC, the field's largest conference — 13,000+ attendees from 111 countries, London 2026.
- Co-organised AAIC Neuroscience Next 2026 Nigeria, an Alzheimer's Association–funded regional conference bringing the field's flagship early-career meeting to West African researchers; second author on the formal evaluation of the Nnewi Hub, now under review at *Alzheimer's & Dementia*.
- Build mentorship and collaboration channels between African neuroscientists and the global dementia research community; authored a published guide for incoming ambassadors so the role outlasts any one holder.

Founder & Lead — Brain-AID, student-led biomedical AI & neuroimaging research community **| 2025 – Present**

- Founded Brain-AID in my penultimate undergraduate year to close a gap I had lived with: my institution had **no neuroimaging analysis training and no research programme at all**. Built it to transfer the training, software skills and conference exposure I had gained through international networks back to students with no other route to them.
- Grown to **350+ members across 74 institutions and 5 countries**, with members going on to present at national and international conferences.
- Now **formally recognised by the College of Medicine, University of Lagos**.

Early Career Representative, Executive Committee — ISTAART Neuroimaging PIA **| Jul 2026 – Present**

- One of **two** early-career representatives on the committee governing **1,870 neuroimaging researchers across 72 countries** — the largest ISTAART community.
- Contribute to AAIC programme planning and to convening the Alzheimer's Imaging Consortium, the PIA's annual scientific meeting; design early-career programming on mentoring, grants and fellowships; serve on the science communication team.

Deep Learning Research Coordinator — CAMERA MRI Africa (SPARK Academy), Montreal (remote) **| Feb 2025 – Present**

- Provide scientific and strategic direction to a 12-member multidisciplinary research team, coordinating experimental design and model implementation for brain tumour segmentation. Certificate.
- As a trainee on the SPARK Academy programme, led a 12-member team developing deep learning segmentation models for adult glioma to a Dice score of **0.73** at MICCAI BraTS-Africa 2024 (Marrakesh), winning a competitive travel award; also competed in the LISA-MICCAI hippocampal segmentation challenge.

Campus Lead, Data Science Nigeria, College of Medicine (2021 – 2023) ·
Reviewer, MICCAI MIRASOL 2026 and NeurIPS Africa in AI 2026

IMPACT & CAPACITY BUILDING

- Built WebMRIQC, free browser-based MRI quality control with DICOM-to-BIDS conversion that eliminates the HPC requirement entirely — **5–10 minutes** per T1 scan over an ordinary internet connection. Serving the entire neuroscience ecosystem and Adopted by the SWiM Program 2026.
- Trained early-career African researchers in neuroimaging software, data management and reproducibility; co-authored the report on CAMERA's programme training 105 African clinicians and students in open neuroimaging analysis.
- Secured ethical approval and curated mammography datasets from **3,325 patients** across six healthcare institutions in Nigeria, Kenya and Uganda; won the MICCAI Open Data Large Grant (2026) to release the data openly, and a ₦6M ACTS Foundation grant for collection and quality assurance.
- Mentored over **150** fellows through a machine learning and analytics curriculum at The Bulb Africa; mentors Nigerian students in the UK through Leverrs.
- Communicates the field to ~5,000 followers on LinkedIn; ISTAART advocacy posts have drawn 18,000+ impressions. Profiled by the University of Lagos as "From Mammograms to Memory Maps: Philip Nkwam's Mission to Humanize AI in African Healthcare."

RESEARCH EXPERIENCE

Visiting Researcher — University College London, Hawkes Institute & Queen Square Institute of Neurology **| Jul – Sep 2026**
Supervisors: Prof David Cash, Prof James Cole, Dr Udunna Anazodo · Funded by UCL Hawkes Institute

- Built an end-to-end low-dose amyloid PET pipeline for an Insight46 cohort subset, reconstructing from raw list-mode detector events with NiftyPET — over 12,000 GPU reconstructions on UCL's HPC cluster — to test how little tracer dose and scan time a reliable amyloid measurement actually needs.
- Quantified the uncertainty of short-frame amyloid imaging through Poisson bootstrap resampling across ~11,000 independent image realisations, and fine-tuned a cross-tracer deep-learning denoising model evaluated against an empirically measured noise floor.
- Invited speaker, UCL Hawkes Seminar — *"African Dementia Research from the Ground Up: Tools Built, Lessons Learned, and What Next."*

Research Assistant — Medical Artificial Intelligence Laboratory (MAILAB), Lagos **| Nov 2024 – Present**

- Designed rs-fMRI preprocessing and cerebrovascular reactivity mapping pipelines for CONNExIN, validated on the PREVENT-AD dataset, and automated MRI quality metrics (SNR, CNR, FD, DVARS) behind WebMRIQC.

Research Assistant & Machine Learning Engineer — St Cyril Cancer Treatment Foundation (with MIT CSAIL) | **Jan 2022 – Dec 2025**

- Led external validation of the MIRAI deep learning breast cancer risk model on African mammograms — testing whether a model trained on Western populations holds where it is most needed — and resolved the metadata gaps that blocked it.

Technical Trainer, Machine Learning & Advanced Analytics Fellowship — The Bulb Africa, Lagos | **Aug 2021 – Aug 2024**

- Designed and delivered a data science curriculum covering machine learning, statistical modelling and applied projects; mentored over 150 fellows through capstone projects and hackathons.

EDUCATION & ACADEMIC DISTINCTIONS

BSc (Hons) Diagnostic Radiography — College of Medicine, University of Lagos, Nigeria | Dec 2018 – Sep 2025

- **Overall Best Graduating Student, Research Category** — three awards: College of Medicine, Navon Pharmaceuticals, and the Alumni Association.
- Final thesis: WebMRIQC, accepted to MICCAI MIRASOL 2026.

AWARDS & GRANTS

ISMRM E.K. Zavoisky Stipend Award (2026) · MICCAI Open Data Large Grant (2026) · ₦6M ACTS Foundation Grant · CAMERA MRI Africa Travel Award (MICCAI, Marrakesh 2024) · AWS BugBust — contributor, Guinness World Record (2021)

PUBLICATIONS & CONFERENCE PRESENTATIONS

Two journal manuscripts under review at Alzheimer's & Dementia; eleven abstracts and papers across ISMRM, AAIC, MICCAI, AORTIC and ARN; first author on four, senior author on one.

Peer-reviewed proceedings

1. Musa T, **Nkwam P**, Sharma A. *Automated Pediatric Brain Hippocampal and Basal Ganglia Segmentation in Ultra-Low Field Magnetic Resonance Images*. Poster, LISA Workshop, MICCAI 2025, South Korea — **LISA-MICCAI journal proceeding**.

Journal manuscripts under review

2. Ogbuagu C, **Nkwam P**, Nyame I, Koech D, Okereke U, Ogbuagu E, Edelmayer R, Nicholson T, Ismail O, Onyike CU, Uwakwe R. ***Decentralizing Dementia Research Capacity in Sub-Saharan Africa: Evaluation of the AAIC Neuroscience Next Nigeria – Nnewi Hub***. Under review, **Alzheimer's & Dementia: The Journal of the Alzheimer's Association** (submitted Sep 2026; ADJ-D-26-03046).
3. Botero-Rodríguez F, Brady JJR, Booncharoen K, Edwards AJ, Egerton R, Fernández Zaionz A, Fornari C, Harput E, Keller G, Mundada NS, Olajide T, Pedersen TJ, Rosales-Leycegui F, Natarajan SG, Sidhu R, Tamayanti WD, Valeriano-Lorenzo E, Vo-Eckerle TT, Watson J, **Nkwam P**, Woodruff SS, Ismail O. ***Investing in the Next Generation: A Call to Action for Early-Career Researchers in Global Dementia Research***. Under review, **Alzheimer's & Dementia: The Journal of the Alzheimer's Association** (submitted Sep 2026; ADJ-D-26-03114).

First-author abstracts & presentations

4. **Nkwam P**, Draper E, Cakmak J, Fajardo A, Kanagasabai K, Tham C, Akinwale O, Aremu O, Donkor Anita N, Umoren C, Raymond C, Dako F, Anazodo U, Mohamed AZ. *Reproducible and Resource-Efficient rs-fMRI Analysis Pipelines for Dementia Research in Low-Resource Settings*. **Oral presentation** at the ISMRM Africa Conference 2025, Tunis, Tunisia.
5. **Nkwam P**, Draper E, Cakmak J, Fajardo A, Kanagasabai K, Tham C, Akinwale O, Umoren C, Aremu O, Donkor Anita N, Raymond C, Dako F, Anazodo U, Mohamed AZ. *Reproducible Neuroimaging Analysis Pipelines for Dementia Research in Resource-Limited Settings: Experience with fMRI Processing and Analysis*. Poster at the Alzheimer's Association International Conference (AAIC 2025), Toronto, Canada, and Lightning Oral Presentation at AAIC Neuroscience NEXT 2026.
6. **Nkwam P**, Oladeji I, Zurakat-Aderibigbe S, Cakmak J, Aduluwa H, Raymond C, Mokua C, Zubair AA, Champanda D, Olusuyi T, Adewole M, Anazodo U. *WebMRIQC: A Web-Based Implementation of MRIQC for Accessible MRI Image Quality Assessment in Resource-Constrained Settings*. MICCAI MIRASOL 2026 — undergraduate final thesis work.
7. **Nkwam P**, Amure I, Adewole M, Isibor V, Gbadamosi Y, Usman A, Shinkafi-Bagudu Z, Animasahun M, Ndoh K. *Improving Breast Cancer Detection in Nigeria with MIRAI Deep Learning Model*. Poster at the 15th AORTIC International Conference on Cancer in Africa 2025, Hammamet, Tunisia (certificate of presentation).

Senior author

8. Ajanaku J, Makinde E, Olufayo H, **Nkwam P**. *Benchmarking Swin-UNETR for brain tumour segmentation on sub-Saharan African datasets: advancing performance in low-resource settings*. ARN Conference 2025, Lagos, Nigeria.
9. Oladeji DA, Adeniyi AT, Anene-Ogbe CM, Babari BMF, Nwasike NC, Ibe CK, Nkenganyi-Aka E, Akinmuleya OI, Anazodo U, Mohamed AZ, Draper E, **Nkwam P**, Fajardo A. *Quantifying Amyloid Burden in Alzheimer's: A Reproducible Centiloid Pipeline for Analysis in Resource-Constrained Settings*. Poster submission at the Alzheimer's Association International Conference (AAIC) 2026 **(supervised)**.

Co-authored

10. Tham C, Aduluwa H, Draper E, Cakmak J, Fajardo A, Akinwale O, Kanagasabai K, **Nkwam P**, Aremu O, Smith CI, Tigbee I, Donkor Anita N, Umoren C, Montalba C, Murithi K, Raymond C, Anazodo U, Mohamed AZ. *Towards Reproducible and Resource-Efficient Perfusion Imaging Analysis for African Dementia Imaging Research*. Poster at the Alzheimer's Association International Conference (AAIC) 2025, Toronto, Canada.
11. Tham C, Draper E, Cakmak JD, Fajardo A, Akinwale O, **Nkwam P**, Aremu O, Smith C, Donkor Anita N, Tigbee I, Umoren C, Raymond C, Anazodo U, Mohamed AZ. *CONNExIN: Advancing Reproducible Perfusion Imaging Analysis for African Neuroscientists*. Poster at the ISMRM Workshop 2025, Spain.
12. Draper EC, Kanagasabai K, Tham C, Akinwale O, Fajardo A, Hough M, Halilibrahimoğlu H, Soula O, Montalba CA, Iyanuoluwa O, García C, Olusuyi T, Solis-Barquero S, Aduluwa HO, Cakmak J, Gallego Rudolf J, **Nkwam PE**, Mohamed AZ, Anazodo U. *Training 105 African clinicians and students in open neuroimaging analysis: CAMERA's dementia research program*. ISMRM Main Conference 2026 submission, Cape Town.
13. Musa T, **Nkwam P**, Sharma A. *Automated Segmentation of Pediatric Hippocampal and Basal Ganglia Structures in Ultra-Low-Field Magnetic Resonance Images*. ISMRM Main Conference 2026 submission, Cape Town.

Profiles: Google Scholar · ResearchGate

SELECTED SPEAKING

Digital Health AI + Data, Birmingham — selected speaker (2024) · DataFest Africa 2024 — Building Optimised AI Models in Low-Resource Settings · DataFest Africa 2021 · Union TV — AI and Youth Development · TechCabal

Projects: Ultrasound breast tumour segmentation (PRECISE 2024) · fMRI preprocessing & CVR mapping (CONNExIN)